

Research Statement

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My research studies how AI systems ground reasoning in perceptual evidence and how their intermediate decisions lead to success or failure. I have investigated these questions in domains with checkable outcomes. I now want to extend this work to hard-to-verify settings, where relevant states are partially observed and an action's value emerges through longer-term consequences.

Grounding reasoning in visual evidence. I believe reasoning should actively direct perception: intermediate deductions should guide what evidence a model examines next. The relevance of a visual detail may become apparent only as an argument develops, motivating a model that revisits the image rather than relying solely on its initial interpretation. In **v1 (COLM 2026)**, I instantiated this view by making proactive visual grounding an explicit operation within the reasoning trajectory: the model learns to point to relevant image regions and incorporate their representations into subsequent reasoning. Disabling pointing at inference reduced performance, showing that actively revisiting visual evidence contributes to reasoning. This provides a foundation for grounding intermediate decisions in evidence when their consequences are difficult to verify.

Diagnosing intermediate decisions. I believe improving reasoning requires understanding where a decision process breaks down, rather than judging only its final outcome. This motivates comparing trajectories at shared states to separate failures that an overall success score conflates. In **WebStep (COLM 2026)**, I instantiated this idea through web environments generated from explicit semantic state models, allowing us to track actions and locate where successful and unsuccessful trajectories diverge. Systems with similar success rates exhibited distinct weaknesses in finding targets and acting on them, revealing different priorities for improvement. The next challenge is to obtain similarly informative feedback when states must be inferred and consequences emerge only over time.

Beyond verifiable tasks. I want to extend these ideas to settings where neither the relevant state nor the value of an action is directly observable. In human-centered applications, a system must infer what matters from incomplete multimodal evidence, while the consequences of its decisions may emerge only over time. I would study how intermediate, evidence-grounded assessments can provide useful learning signals in such settings, and when optimizing them actually improves longer-term outcomes. The broader goal is to move from reasoning and diagnosis under explicit supervision to learning useful behavior when feedback is partial, delayed, and difficult to verify.

Working with Diyi Yang, Emma Brunskill, and domain experts would provide an ideal setting to pursue this direction, combining multimodal understanding with sequential decision-making and rigorous evaluation of long-term effects. I hope to develop AI systems that can reason from complex human evidence while learning from consequences that cannot be reduced to immediate, easily verifiable rewards.